

— IE437 · DATA-DRIVEN DECISION MAKING AND CONTROL — LECTURE 2

# Fundamentals on Bayesian Statistics

*Don't pick a number — carry a belief*

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The first crossing — from a world handed to us to a world inferred from data

— HANDOFF

# The handoff — the objective was uncertain

Lecture 1 minimised a *known*  $f$ . In practice  $f$  is fitted to noisy data — and its parameters are uncertain.

# Where we are — the model stops being exact

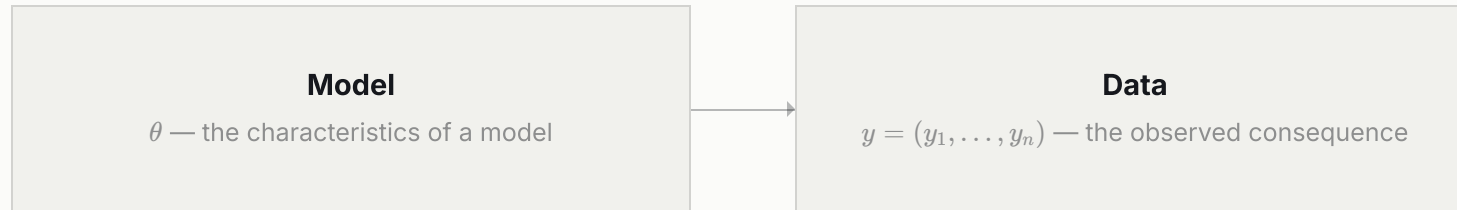
|        |              |                  |                                 |        |                    |
|--------|--------------|------------------|---------------------------------|--------|--------------------|
| STAGES | static ● — ○ | MODEL · CROSSING | model-based → data-driven ○ — ● | AGENTS | single agent ● — ○ |
|--------|--------------|------------------|---------------------------------|--------|--------------------|

| MODEL-BASED (CERTAIN) |                      | DATA-DRIVEN (UNCERTAIN)     |
|-----------------------|----------------------|-----------------------------|
| Static, single        | optimisation (Lec 1) | Bayesian statistics (Lec 2) |

Lecture 1 minimised a *known*  $f$ . But in practice  $f$  comes from a [model fit to noisy data](#) — its parameters are uncertain. A single "best" decision then rests on a single, possibly wrong, parameter estimate.

So this lecture steps back from *deciding* to *modelling under uncertainty*: how to represent, and update, what we believe about a model's parameters given data. It is the first encounter with the course's central axis — [from a world handed to us, to a world inferred from data](#).

# Statistics infers the cause that generated the data



For a tossed coin:  $\theta$  is the probability of a head; the data is the sequence *(Head, Head, Tail, ...)*. The arrow runs left to right in the world and **right to left in statistics** — we see consequences and must infer the cause.

## TWO VIEWS ON PROBABILITY

- **Frequentist** — probability has meaning only as the limiting case of *repeated measurements*; it is a frequency of events.
- **Bayesian** — probability is a *degree of certainty* about a statement; it is about our own knowledge of an event.

## TWO VIEWS ON STATISTICS

- **Frequentist** — data is a repeatable random sample; **parameters are fixed and unchanging**.
- **Bayesian** — data is fixed, observed from the realised sample; **parameters are unknown and described probabilistically**.

# The coin, and what each view struggles with

## FREQUENTIST

$\theta$  is the relative frequency of heads in a "large number" of "identical flips". Nothing matters more than repeatability.

**The trouble.** With small  $n$  the estimate is wildly unstable —

$$\frac{\# \text{Success}}{\# \text{Trials}} = \frac{1}{3}, \frac{5}{6}, \frac{5}{13}, \frac{129}{313}, \frac{61423}{123400}$$

and an "identical flip" is not a thing that exists.

## BAYESIAN

$\theta$  is itself uncertain. Express the belief as a distribution  $p(\theta)$  — each  $\theta$  can correspond to a different orientation, force, surface.

**The trouble.**  $p(\theta)$  is **subjective**. How do you specify it, and how much does the answer depend on it?

Act 2 answers the Bayesian objection quantitatively: the prior's influence is not a matter of taste but a *weight*, and that weight is  $\beta/(\beta + n)$  — it vanishes as data accumulates.

# The thesis — a belief, not a guess

When you don't know a parameter, don't pick one value — [carry a distribution over all of them](#).

|                        | FREQUENTIST                       | BAYESIAN                               |
|------------------------|-----------------------------------|----------------------------------------|
| the parameter $\theta$ | fixed but unknown                 | a random variable                      |
| what you estimate      | by maximising the likelihood      | a full posterior distribution          |
| what you get           | one point estimate $\hat{\theta}$ | a balance of prior belief and evidence |

The Bayesian move — treating the unknown itself as a distribution that data sharpens — is the engine behind Gaussian processes (Ch 4), surrogate uncertainty (Ch 5), and belief-state decision making (Ch 3 onward). Learn it once, reuse it all term.

# The roadmap — four questions



- **Q1 — Why represent belief as a distribution?** Bayes' rule and the prior–posterior picture.
- **Q2 — How does data update belief?** The posterior as a **balance** of prior and evidence; conjugacy.
- **Q3 — What if I must commit to one value?** MLE vs. MAP — and why **regularisation is a prior**.
- **Q4 — How do I predict with my uncertainty intact?** The **posterior predictive** distribution.

— ACT 1

# Belief as a distribution

**Q1.** One line of algebra that turns a belief and a measurement into a new belief.



# Bayes' rule — the whole of inference on one line

Q1 Why a distribution?

Q2 How does data update it?

Q3 Must I commit?

Q4 How do I predict?

## The posterior

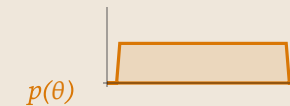
the probability of the parameter  $\theta$  given the evidence (data)



$$p(\theta|data) = \frac{p(data|\theta)p(\theta)}{p(data)}$$

## The prior

the belief about  $\theta$  before any data



## The likelihood

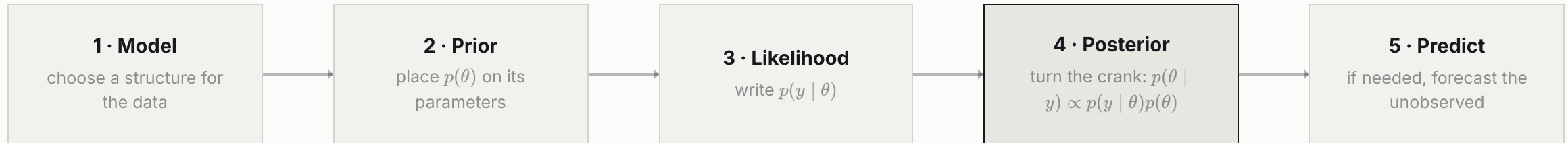
how probable this data is under each candidate  $\theta$  — the only place the measurement enters

## The evidence — a normaliser

$p(data) = \int p(data|\theta) p(\theta) d\theta$ , the probability of the data over every possibility. It does not depend on  $\theta$ , which is why **posterior**  $\propto$  **likelihood**  $\times$  **prior** is usually enough

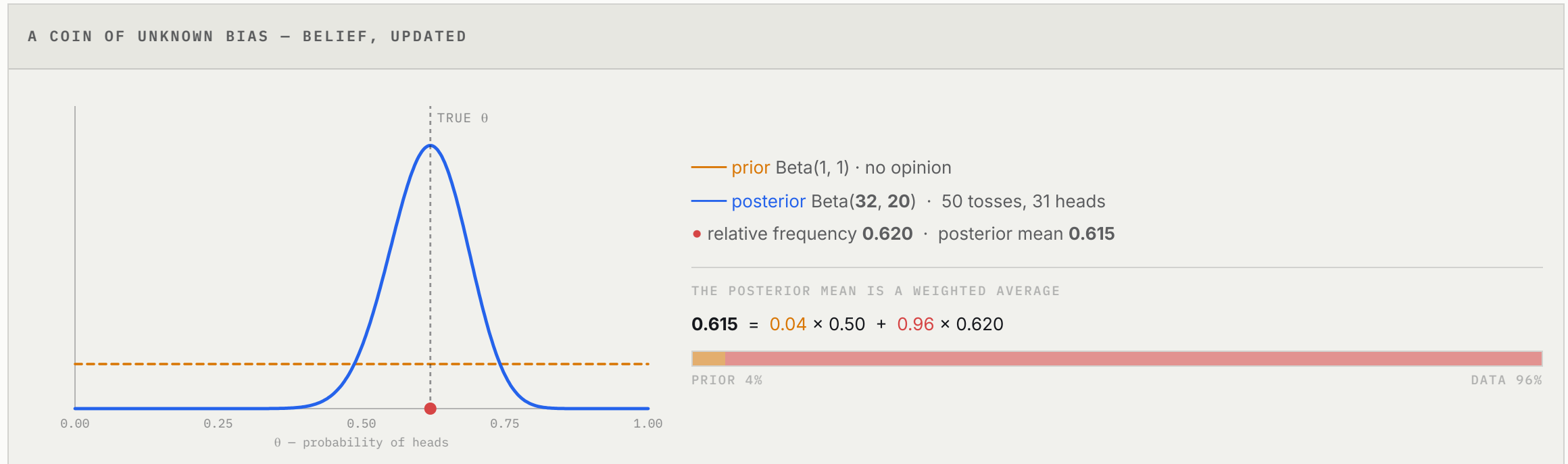
Belief in = prior; belief out = posterior; **the data does the turning.**

# The procedure, not the formula



Bayes' rule is not a formula to memorise but a **procedure**. Every lecture in Part II is this same five-step loop with a different model in step 1 — a Bernoulli here, a graph in Ch 3, a whole function in Ch 4.

# The picture — belief sharpening with evidence

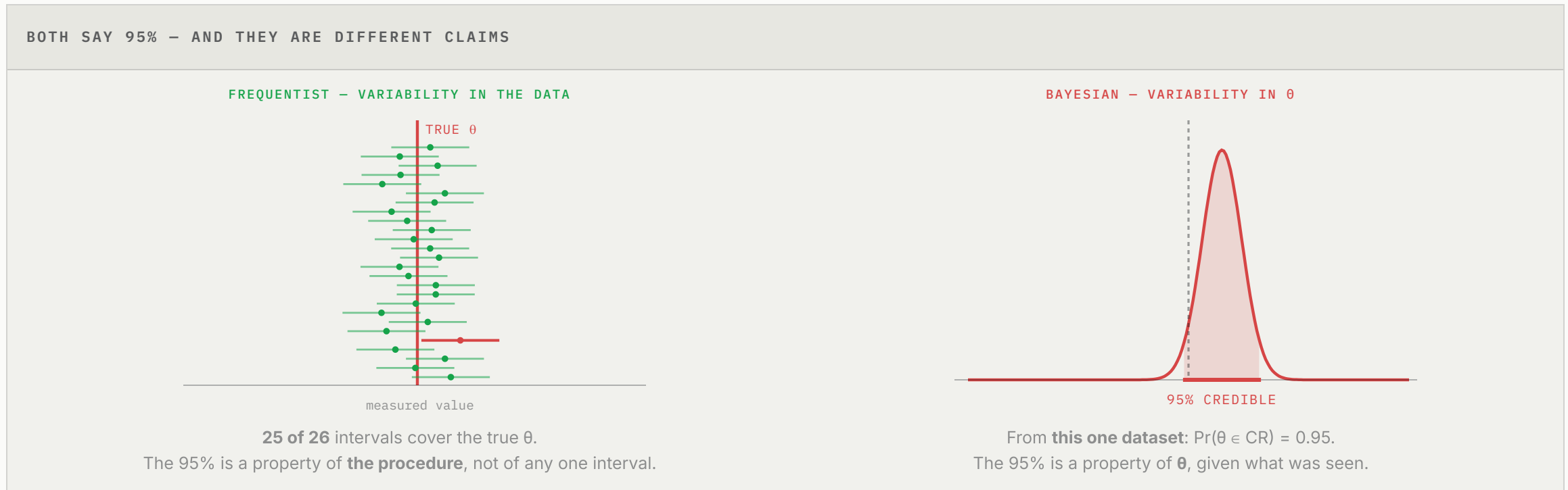


A flat prior over a coin's bias; each toss reshapes it. Watch epistemic uncertainty shrink in front of you — and read the two weights below the chart: the posterior mean is always **a weighted average of the prior mean and the data's estimate**. That is Act 2, arrived at early.

# What a distribution buys that a point cannot

Early on the posterior is wide — and an honest decision-maker acts cautiously. That "width" becomes the explore/exploit signal of Bayesian optimisation (Ch 4), the uncertainty penalty of Ch 5, and the belief state of every sequential method after that.

It also changes what a "95% interval" *means*, and the two meanings are not interchangeable:



— ACT 2

# How data updates belief

**Q2.** The update is not a matter of taste. It is a weighted average, and the weights are explicit.

# The coin, done both ways

Q1 Why a distribution?

Q2 How does data update it?

Q3 Must I commit?

Q4 How do I predict?

## FREQUENTIST — MAXIMUM LIKELIHOOD

With  $Y_i \sim B(\theta)$  the likelihood of  $n$  tosses is

$$L(\theta) = p(y_1, \dots, y_n \mid \theta) = \theta^{\sum y_i} (1 - \theta)^{n - \sum y_i}$$

Setting  $dL/d\theta = 0$  gives

$$\hat{\theta}_{\text{ML}} = \frac{\sum y_i}{n}$$

**MLE returns the relative frequency.**  $\sum y_i$  is a *sufficient statistic*: the order of the tosses never mattered.

## BAYESIAN — CONJUGATE UPDATE

Likelihood  $Y \sim \text{Bin}(n, \theta)$ , prior  $\theta \sim \text{Beta}(\alpha, \beta)$ :

$$\begin{aligned} p(\theta \mid y) &\propto \theta^y (1 - \theta)^{n-y} \cdot \theta^{\alpha-1} (1 - \theta)^{\beta-1} \\ &= \text{Beta}(\theta \mid \alpha + y, \beta + n - y) \end{aligned}$$

$\alpha$  is a **pseudo-count of successes**,  $\beta$  a pseudo-count of failures. The prior is simply data you already believed you had.

# The posterior is a balance — prior vs. evidence

$$\mathbb{E}[\theta \mid y] = \frac{\alpha + y}{\alpha + \beta + n} = \underbrace{\frac{\alpha + \beta}{\alpha + \beta + n}}_{\text{weight on prior}} \mathbb{E}[\theta] + \underbrace{\frac{n}{\alpha + \beta + n}}_{\text{weight on data}} \hat{\theta}_{\text{ML}}$$

The posterior mean is a **weighted average** of the prior mean and the data's estimate.

- small  $n$  — the *prior* dominates; belief barely moves;
- large  $n$  — the *data* dominates, the prior washes out and  $\mathbb{E}[\theta \mid y] \rightarrow \hat{\theta}_{\text{ML}}$ ;
- $\alpha + \beta$  tunes the **strength** of the prior — how many pseudo-observations it is worth.

# Two laws that say the same thing, exactly

THE PRIOR MEAN IS THE AVERAGE OF EVERY POSTERIOR MEAN

$$\mathbb{E}[\theta] = \mathbb{E}[\mathbb{E}[\theta \mid y]]$$

Before seeing data, your belief is already the average of every belief you might end up with.

THE POSTERIOR IS ON AVERAGE TIGHTER THAN THE PRIOR

$$\text{Var}(\theta) = \mathbb{E}[\text{Var}(\theta \mid y)] + \text{Var}(\mathbb{E}[\theta \mid y])$$

The posterior variance is smaller than the prior variance by exactly the variation in posterior means across possible datasets.

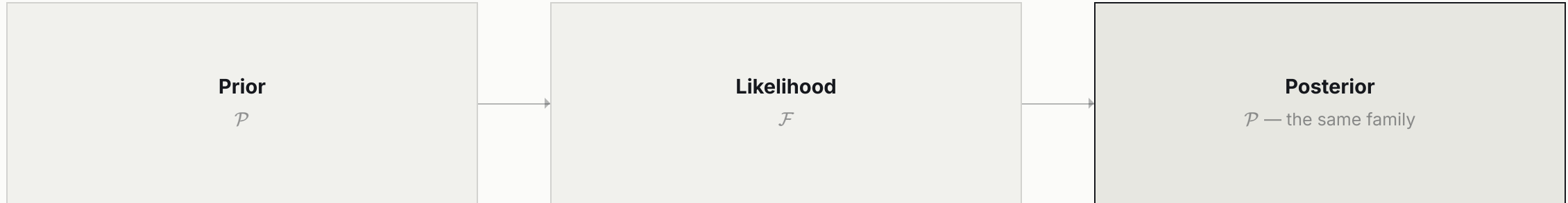
**Data cannot, on average, make you less certain.**

The posterior sits at a compromise between prior information and data, and the compromise is controlled more and more by the data as the sample grows.



# Conjugacy — why any of this is computable

A prior is **conjugate** to a likelihood when the posterior lands in the same family as the prior. The update then has a closed form and there is no integral to evaluate.



- you know the posterior's form in advance, so mean, mode and variance are immediate;
- the prior has a readable meaning — a Beta prior is just adding pseudo-counts;
- and the evidence  $p(y) = \int p(y | \theta)p(\theta) d\theta$  can actually be carried out.

# The conjugate pairs worth knowing

| LIKELIHOOD                               | CONJUGATE PRIOR | POSTERIOR     |
|------------------------------------------|-----------------|---------------|
| Binomial · Negative binomial · Geometric | Beta            | Beta          |
| Poisson · Exponential                    | Gamma           | Gamma         |
| Normal, mean unknown                     | Normal          | Normal        |
| Normal, variance unknown                 | Inverse Gamma   | Inverse Gamma |
| Normal, both unknown                     | Normal–Gamma    | Normal–Gamma  |
| Multinomial                              | Dirichlet       | Dirichlet     |

In every case the posterior parameters are the prior's **plus a sufficient statistic of the data** — which is exactly why the posterior mean is always a prior-vs-data weighted average, and why a prior behaves like a stock of pseudo-observations.

# A worked example — counting Pokémon

Example 2.1 · Poisson likelihood, Gamma prior

Counts in 20 districts of San Francisco: 14, 13, 7, 10, 15, 15, 2, 13, 13, 11, 10, 13, 5, 13, 9, 12, 9, 12, 8, 7.

## THE MODEL

$Y_i \sim \text{Poisson}(\lambda)$ , and a Gamma prior with mean 20 and standard deviation 10:

$$\mathbb{E}[\lambda] = \frac{\alpha}{\beta} = 20, \quad \text{Var}(\lambda) = \frac{\alpha}{\beta^2} = 10^2 \Rightarrow \alpha = 4, \beta = 0.2$$

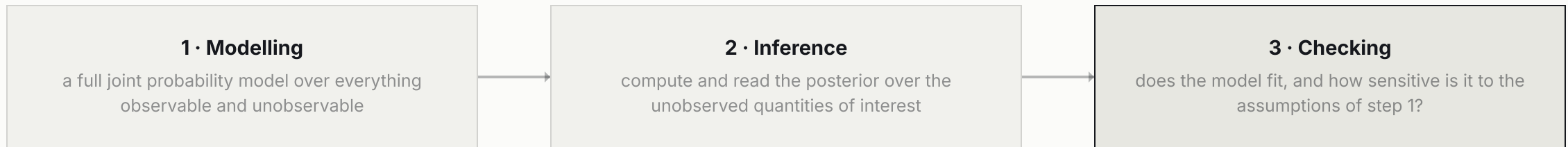
## THE UPDATE

$$p(\lambda \mid y) = \text{Gamma}(\alpha + n\bar{y}, \beta + n) = \text{Gamma}(215, 20.2)$$

$$\mathbb{E}[\lambda \mid y] = \frac{215}{20.2} = 10.64, \quad \text{Var} = \frac{215}{20.2^2} = 0.527$$

The prior said 20 and the data says 10.55. The posterior says **10.64** — the twenty observations have all but erased a prior worth  $\beta = 0.2$  pseudo-observations. The same weighted average, in numbers.

# Three steps, and a loop



Step 3 loops back to step 1. A Bayesian analysis is not finished when the posterior is computed — it is finished when the model has survived being doubted.

— ACT 3

# **Collapsing to a point: MLE, MAP, regularisation**

**Q3.** Sometimes one number is required. There are two principled ways to extract one — and one of them has a familiar face.

# When you must commit — MLE vs. MAP

Q1 Why a distribution?

Q2 How does data update it?

Q3 Must I commit?

Q4 How do I predict?

## MAXIMUM LIKELIHOOD

$$\hat{\theta} = \arg \max_{\theta} p(y \mid \theta)$$

Ignores the prior. The frequentist's point estimate — and, as the coin showed, the relative frequency.

## MAXIMUM A POSTERIORI

$$\hat{\theta} = \arg \max_{\theta} p(\theta \mid y) \propto \arg \max_{\theta} p(y \mid \theta) p(\theta)$$

The peak of the posterior. MLE **plus a prior term**.

With abundant data the two coincide — the likelihood swamps the prior. With scarce data the prior is what keeps the estimate sane.

And that prior term has **a familiar face**.

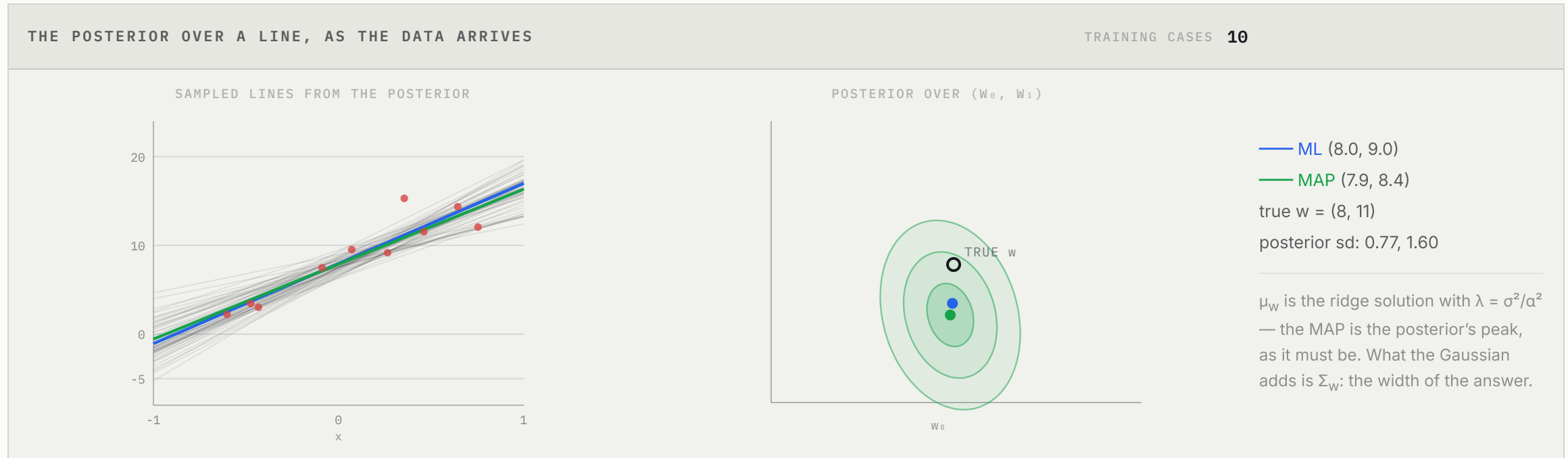
# Linear regression, and the same fit four times

The rest of this act is one worked problem — fit a line — approached from four directions that turn out to be the same direction.

| # | STARTING POINT                                                                        | RESULT                                                                                       |
|---|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 1 | <b>Optimisation.</b> $\min_w \frac{1}{2} \ y - \mathbf{X}w\ _2^2$                     | normal equation $\hat{w} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top y$              |
| 2 | <b>MLE.</b> $y_i = w^\top x_i + \epsilon_i, \epsilon_i \sim \mathcal{N}(0, \sigma^2)$ | $\log p(y \mid \mathbf{X}, w) = \text{const} - \frac{1}{\sigma^2} J(w)$ — the same $\hat{w}$ |
| 3 | <b>MAP.</b> add a Gaussian prior on $w$                                               | ridge, $\hat{w} = (\mathbf{X}^\top \mathbf{X} + \lambda I)^{-1} \mathbf{X}^\top y$           |
| 4 | <b>Full Bayes.</b> keep the whole posterior over $w$                                  | a predictive <i>distribution</i> , with error bars                                           |

Maximising the log likelihood with respect to  $w$  **is** minimising the squared error. Least squares was never a separate idea from maximum likelihood; it is Gaussian-noise MLE wearing different clothes.

# The Bayesian fit, as the data arrives



Draw  $w \sim p(w \mid \mathbf{X}, y) = \mathcal{N}(\mu_w, \Sigma_w)$  and plot the line. With two points the posterior is broad and the sampled lines fan out; with a hundred it has collapsed onto one answer and ML and MAP agree. **What full Bayes keeps and the point estimates throw away is exactly that fan.**

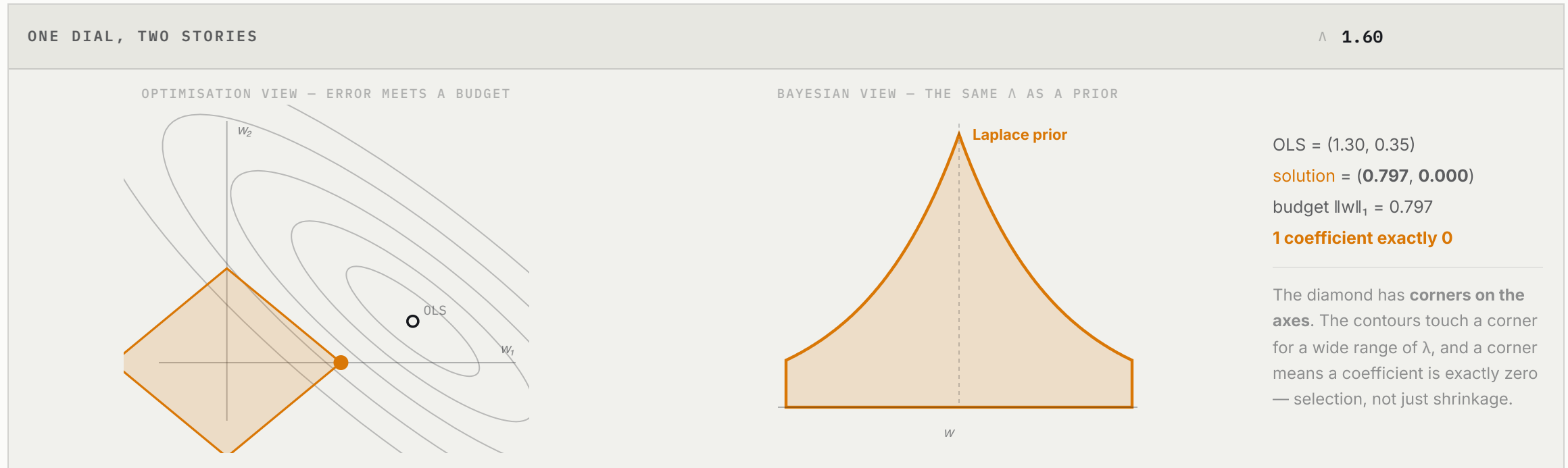


# The unification — regularisation **is** a prior

| ESTIMATOR              | OPTIMISATION VIEW                                    | BAYESIAN VIEW (MAP WITH A PRIOR)        |
|------------------------|------------------------------------------------------|-----------------------------------------|
| ordinary least squares | $\min_w \ y - \mathbf{X}w\ _2^2$                     | MLE, Gaussian noise, no prior           |
| Ridge                  | $\min_w \ y - \mathbf{X}w\ _2^2 + \lambda \ w\ _2^2$ | MAP with a <b>Gaussian</b> prior on $w$ |
| Lasso                  | $\min_w \ y - \mathbf{X}w\ _2^2 + \lambda \ w\ _1$   | MAP with a <b>Laplace</b> prior on $w$  |

| THE PUNCHLINE                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A regulariser is not an ad-hoc penalty — it is <b>the log of a prior</b> . Ridge's $\ell_2$ term is a Gaussian prior; Lasso's $\ell_1$ term is a Laplace prior, whose sharp peak at zero is what produces sparsity. |

# Two views of the same picture



On the left, the optimisation view: level sets of the squared error meeting a budget set — a ball for  $\ell_2$ , a diamond for  $\ell_1$ , whose corners are why Lasso zeroes coefficients. On the right, the Bayesian view: the very same  $\lambda$  read as **the width of a prior**. Turn the dial and watch both stories move together.

This is the lecture's deepest bridge: **the optimisation world and the Bayesian world are the same world**, seen through different lenses. "Add a penalty to avoid overfitting" and "encode a prior belief about the weights" are one act.

— ACT 4

# Predicting with uncertainty intact

**Q4.** A point estimate predicts with the best  $\theta$ . Full Bayes predicts with all of them.

# Full Bayes — integrate, don't plug in

Q1 Why a distribution?

Q2 How does data update it?

Q3 Must I commit?

Q4 How do I predict?

MLE and MAP collapse the posterior to a point and then predict, discarding everything else. The fully Bayesian prediction **averages over the entire posterior**:

$$p(\hat{y} | y) = \int p(\hat{y} | \theta) \underbrace{p(\theta | y)}_{\text{posterior}} d\theta$$

Don't predict with the *best*  $\theta$  — predict with **all  $\theta$ , weighted by belief**.

# What the integral buys, and what it costs

- it **propagates uncertainty** into the prediction: the predictive distribution is wider exactly when the posterior is;
- it guards against the overconfidence of a single fitted parameter;
- the price is the integral — closed form for conjugate models, otherwise approximated by **sampling** (MCMC) or variational methods.

And for conjugate pairs the answer is a named distribution, every time:

| MODEL                | PRIOR PREDICTIVE $P(Y)$                            | POSTERIOR PREDICTIVE $P(Y   Y)$                                  |
|----------------------|----------------------------------------------------|------------------------------------------------------------------|
| <b>Binomial–Beta</b> | Beta–Binomial( $y \mid n, \alpha, \beta$ )         | Beta–Binomial( $\hat{y} \mid n, \alpha + y, \beta + n - y$ )     |
| <b>Poisson–Gamma</b> | Negative binomial( $y \mid \alpha, \beta$ )        | Negative binomial( $\hat{y} \mid \alpha + n\bar{y}, \beta + n$ ) |
| <b>Normal–Normal</b> | $\mathcal{N}(y \mid \mu_0, \sigma_Y^2 + \tau_0^2)$ | $\mathcal{N}(\hat{y} \mid \mu_1, \sigma_Y^2 + \tau_1^2)$         |

# The Normal case, read as precision

For  $Y_i \sim \mathcal{N}(\theta, \sigma_Y^2)$  with  $\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$ , the posterior is  $\mathcal{N}(\mu_1, \tau_1^2)$  with

$$\mu_1 = \frac{\frac{\mu_0}{\tau_0^2} + \frac{n\bar{y}}{\sigma_Y^2}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma_Y^2}}, \quad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma_Y^2}$$

**Precisions add.** The posterior mean is the prior mean and the data mean weighted by their precisions; the posterior precision is the prior's plus the data's. Sharpen the prior ( $\tau_0 \downarrow$ ) and it pulls harder; sharpen the measurements ( $\sigma_Y \downarrow$ ) or take more of them ( $n \uparrow$ ) and the data pulls harder.

And the predictive variance is  $\sigma_Y^2 + \tau_1^2$  — measurement noise you can never remove, plus parameter uncertainty you can. The two kinds of uncertainty Lecture 0 named, arriving as two separate terms in one formula. When the "parameter" is an entire function, this same predictive integral becomes Gaussian-process regression — the core of Lecture 4.

— CLOSING

# Closing

Belief established. Now: what if there are many unknowns, and they interact?

# Where we are — belief established, complexity ahead

| MODEL-BASED    |                                 | DATA-DRIVEN                                                      |
|----------------|---------------------------------|------------------------------------------------------------------|
| Static, single | optimisation ( <i>Lec 1 ✓</i> ) | <a href="#">Bayesian statistics</a> ( <i>Lec 2 ✓</i> ) → Lec 3–4 |

We can now hold and update a belief over a parameter, collapse it when forced (MLE / MAP), and predict with it intact (full Bayes). But we modelled belief over [one parameter \(vector\)](#).

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Real systems have *many* interacting random variables — failures that cause failures, symptoms that imply causes. A joint distribution over them is exponentially large. How do we represent, and reason within, belief over a whole system? That is Lecture 3: [a belief about many things is a graph](#).



Frequentists estimate a number; Bayesians carry a  
distribution.

A prior turned by data into a posterior, collapsed to a point only when forced — and even then, a prior in  
disguise — and integrated over when prediction must stay honest.

# Questions?

The single idea — *the unknown is itself a distribution that data sharpens* — is the seed of everything data-driven this term. Regularisation was a prior; a Gaussian process will be a prior over functions; a belief state will be a prior carried through time.

— APPENDIX

# Backup slides

Complete derivations, kept out of the narrative.

# Backup 1 — the Beta–Binomial pipeline, end to end

**Likelihood.**  $Y \sim \text{Bin}(n, \theta)$ , so  $p(y \mid \theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$ . **Prior.**  $\theta \sim \text{Beta}(\alpha, \beta)$ , so  $p(\theta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$ .

**Posterior.** Drop every factor free of  $\theta$ :

$$p(\theta \mid y) \propto \theta^y (1 - \theta)^{n-y} \theta^{\alpha-1} (1 - \theta)^{\beta-1} = \theta^{\alpha+y-1} (1 - \theta)^{\beta+n-y-1} = \text{Beta}(\alpha + y, \beta + n - y)$$

**Prior predictive.** Integrating the likelihood against the prior and collecting Gamma functions gives  $p(y) = \text{Beta–Binomial}(y \mid n, \alpha, \beta)$  — a binomial whose success probability is itself random and Beta-distributed.

**Posterior predictive.** The same integral with  $\alpha, \beta$  replaced by their updated values:  $p(\hat{y} \mid y) = \text{Beta–Binomial}(\hat{y} \mid n, \alpha + y, \beta + n - y)$ .

**Moments.**  $\mathbb{E}[\theta \mid y] = \frac{\alpha + y}{\alpha + \beta + n}$  and  $\text{Var}(\theta \mid y) = \frac{\mathbb{E}[\theta \mid y](1 - \mathbb{E}[\theta \mid y])}{\alpha + \beta + n + 1}$ , so as  $n \rightarrow \infty$  the variance falls like  $p(1 - p)/n$  — the familiar frequentist rate, recovered.

## Backup 2 — Lasso as MAP with a Laplace prior

Gaussian likelihood  $p(y \mid \mathbf{X}, w) = \prod_i \mathcal{N}(y_i \mid w^\top x_i, \sigma^2)$  and Laplace prior  $p(w) = \prod_k \frac{\lambda}{2\sqrt{\tau^2}} \exp\left(-\frac{\lambda}{\sqrt{\tau^2}}|w_k|\right)$ .

Posterior  $\propto$  likelihood  $\times$  prior; take logs:

$$\log p(w \mid \mathbf{X}, y) = -\frac{1}{2\sigma^2} \sum_i (y_i - w^\top x_i)^2 - \frac{\lambda}{\sqrt{\tau^2}} \sum_k |w_k| + \text{const}$$

Maximising over  $w$  is therefore

$$\hat{w} = \arg \max_w \log p(w \mid \mathbf{X}, y) = \arg \min_w \|y - \mathbf{X}w\|_2^2 + \lambda_1 \|w\|_1 \quad \blacksquare$$

which is exactly Lasso; a Gaussian prior gives the  $\ell_2$  (ridge) penalty by the identical argument.

**Why  $\ell_1$  is sparse.** The Laplace prior puts a sharp peak of mass at zero, so the MAP drives weak coefficients exactly to 0 — automatic feature selection, arrived at from a belief rather than a heuristic.

## Backup 3 — the Bayesian posterior over regression weights

With likelihood  $p(y \mid w) = \frac{1}{(2\pi\sigma^2)^{m/2}} \exp\left(-\frac{1}{2\sigma^2} \sum_i (y_i - w^\top x_i)^2\right)$  and prior  $p(w) = \mathcal{N}(w \mid \mathbf{0}, \alpha^2 I)$ , the log posterior collects into a quadratic in  $w$ :

$$\log p(w \mid \mathbf{X}, y) = -\frac{1}{2\sigma^2} y^\top y + \frac{1}{\sigma^2} y^\top \mathbf{X}w - \frac{1}{2} w^\top \left[ \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X} + \frac{1}{\alpha^2} I \right] w + \text{const}$$

A quadratic log density is a Gaussian, so

$$p(w \mid \mathbf{X}, y) = \mathcal{N}(w \mid \mu_w, \Sigma_w), \quad \Sigma_w = \left[ \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X} + \frac{1}{\alpha^2} I \right]^{-1}, \quad \mu_w = \Sigma_w \left( \frac{1}{\sigma^2} \mathbf{X}^\top y \right)$$

$\mu_w$  is the ridge solution with  $\lambda = \sigma^2/\alpha^2$  — the MAP is the posterior's peak, as it must be. What the Gaussian adds is  $\Sigma_w$ : the *width* of the answer, which is what the sampled lines in the Act 3 widget are drawn from, and what Lecture 4 will use to decide where to sample next.